

DAS DIAL PROBLEM STATEMENT 4

Title	DAS Error Pattern Analyzer
Industry Mentors	Ms Soofrina Mubarak Ms Kalphna C.
Project Context	Dyslexia Association of Singapore (DAS) supports students with dyslexia and other learning differences through specialised literacy intervention programmes. Students with dyslexia and related learning differences often display recurring spelling and writing error patterns. Identifying these patterns manually requires significant educator expertise and time. DAS is exploring an NLP-assisted Error Pattern Analyzer capable of automatically detecting and categorising literacy-related writing errors from student work samples.
Project Objectives	How can Natural Language Processing (NLP) and AI techniques be utilised to: ● Analyse student writing samples ● Detect recurring spelling and literacy-related error patterns ● Categorise errors (e.g., phonological, orthographic, morphological) ● Generate interpretable diagnostic summaries for educators ● Support data-informed intervention planning

Project Deliverables	Presentation with demo of prototype Prototype with the following technical requirements: • NLP-enabled web application • Text ingestion and analysis pipeline • Error classification and tagging system • Database for storing writing samples and analytics • Dashboard for visualising learner error trends • Test cases and evaluation metrics Documentation • NLP architecture and tokenisation strategy • Error taxonomy and classification rationale • AI evaluation methodology • Project management methodology • User testing findings
Main form of expected project deliverables	An AI-assisted student writing diagnostic platform
Crucial properties to test for the delivered software	Accuracy, Interpretability, Functionality, Security, Reliability
Additional related resources	www.das.org.sg